

EDEXCEL INTERNATIONAL A LEVEL

# WMA12/01 October 2022 Worked Solutions

October 2022 paper rebuilt with the worked solution after each question.

**10**

paper questions

**WMA12**

Pure 2

**Single-  
paper  
solutions**standalone IAL  
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - [eliteigcse.com](https://eliteigcse.com)**P2**

**PAST PAPER**

# WMA12/01 October 2022

October 2022 | 10 questions | 75 marks

**10**

questions

**75**

marks

**Answers**

worked solution  
after each  
question

## Question 1

Proof

1. Given that  $a$ ,  $b$  and  $c$  are integers greater than 0 such that

- $c = b + 2$
- $a + b + c = 10$

Prove, by exhaustion, that the product of  $a$ ,  $b$  and  $c$  is always even.

You may use the table below to illustrate your answer.

(3)

*You may not need to use all rows of this table.*

| $a$ | $b$ | $c$ |
|-----|-----|-----|
|     |     |     |

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(Total for Question 1 is 3 marks)

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(Total for Question 1 is 3 marks)

# Worked Solution - Question 1

Topic group

## 1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

## 2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

## 3. State the final result

Write the answer in the form requested by the question: is exhaustive and concludes

### Final answer

is exhaustive and concludes

## Question 2

## Binomial Expansion

2. A curve  $C$  has equation  $y = f(x)$  where

$$f(x) = (2 - kx)^5$$

and  $k$  is a constant.

Given that when  $f(x)$  is divided by  $(4x - 5)$  the remainder is  $\frac{243}{32}$

- (a) show that  $k = \frac{2}{5}$

(2)

- (b) Find the first three terms, in ascending powers of  $x$ , of the binomial expansion of

$$\left(2 - \frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Using the solution to part (b) and making your method clear,

- (c) find the gradient of  $C$  at the point where  $x = 0$

(2)

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(Total for Question 2 is 7 marks)

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(Total for Question 2 is 7 marks)

# Worked Solution - Question 2

## 1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

## 2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

## 3. State the final result

Write the answer in the form requested by the question:  $- + + - = f 32 = -$

### Final answer

$$- + + - = f 32 = -$$

## Question 3

## Trigonometric Equations

3. A sequence  $a_1, a_2, a_3, \dots$  is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i)  $a_1$

(ii)  $a_2$

(iii)  $a_3$

(3)

- (b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

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(Total for Question 3 is 7 marks)

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(Total for Question 3 is 7 marks)

# Worked Solution - Question 3

## 1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

## 2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

## 3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

### Final answer

The required value or expression is obtained by the final simplification.



## Question 4

## Sequences &amp; Series

4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal,  $w$  kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \qquad 2 \leq t \leq 18$$

where  $t$  is the age of the mammal in months and  $a$  is a constant.

Given that the weight of the mammal was 10 kg when  $t = 3$

- (a) show that  $a = 1.072$  correct to 3 decimal places. (3)

Using  $a = 1.072$

- (b) find an equation for  $t$  in terms of  $w$  (3)

- (c) find the value of  $t$  when  $w = 15$ , giving your answer to 3 significant figures. (2)

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(Total for Question 4 is 8 marks)

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(Total for Question 4 is 8 marks)

# Worked Solution - Question 4

**1. Translate the sequence rule**

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

**2. Find the requested term or sum**

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

**3. State the final result**

Write the answer in the form requested by the question:  $\times =$

**Final answer** $\times =$

## Question 5

## Trigonometric Equations

5.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$(3 \cos \theta - \tan \theta) \cos \theta = 2$$

can be written as

$$3 \sin^2 \theta + \sin \theta - 1 = 0 \quad (3)$$

(b) Hence solve for  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ 

$$(3 \cos 2x - \tan 2x) \cos 2x = 2 \quad (5)$$

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(Total for Question 5 is 8 marks)

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(Total for Question 5 is 8 marks)

# Worked Solution - Question 5

## 1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

## 2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

## 3. State the final result

Write the answer in the form requested by the question:  $\theta = \dots = 1$

### Final answer

$$\theta = \dots = 1$$

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7 marks

## Question 6

Integration

6. The curve  $C_1$  has equation  $y = f(x)$ .

A table of values of  $x$  and  $y$  for  $y = f(x)$  is shown below, with the  $y$  values rounded to 4 decimal places where appropriate.

|     |   |        |     |        |      |
|-----|---|--------|-----|--------|------|
| $x$ | 0 | 0.5    | 1   | 1.5    | 2    |
| $y$ | 3 | 2.6833 | 2.4 | 2.1466 | 1.92 |

- (a) Use the trapezium rule with all the values of  $y$  in the table to find an approximation for

$$\int_0^2 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

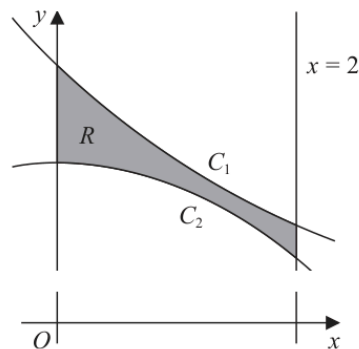


Figure 1

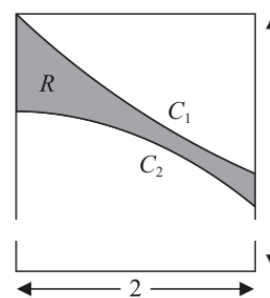


Figure 2

The region  $R$ , shown shaded in Figure 1, is bounded by

- the curve  $C_1$
- the curve  $C_2$  with equation  $y = 2 - \frac{1}{4}x^2$
- the line with equation  $x = 2$
- the  $y$ -axis

The region  $R$  forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region  $R$  inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

- (b) calculate an estimate for the percentage of the logo which is shaded.

(4)

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(Total for Question 6 is 7 marks)

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(Total for Question 6 is 7 marks)

# Worked Solution - Question 6

## 1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

## 2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

## 3. State the final result

Write the answer in the form requested by the question: ( ) 25

### Final answer

( ) 25

## Question 7

## Integration

7. The curve  $C$  has equation

$$y = \frac{12x^3(x-7) + 14x(13x-15)}{21\sqrt{x}} \quad x > 0$$

- (a) Write the equation of  $C$  in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where  $a$ ,  $b$ ,  $c$  and  $d$  are fully simplified constants.

(3)

The curve  $C$  has three turning points.

Using calculus,

- (b) show that the  $x$  coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Given that the  $x$  coordinate of one of the turning points is 1

- (c) find, using algebra, the exact  $x$  coordinates of the other two turning points.

*(Solutions based entirely on calculator technology are not acceptable.)*

(3)

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(Total for Question 7 is 9 marks)

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(Total for Question 7 is 9 marks)



# Worked Solution - Question 7

## 1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

## 2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

## 3. State the final result

Write the answer in the form requested by the question:  $\pm 2.85 \times 10^{-4} \times \pm =$

### Final answer

$$\pm 2.85 \times 10^{-4} \times \pm =$$

## Question 8

## Geometric Sequences

8. A geometric sequence has first term  $a$  and common ratio  $r$

Given that  $S_{\infty} = 3a$

- (a) show that  $r = \frac{2}{3}$

(2)

Given also that

$$u_2 - u_4 = 16$$

where  $u_k$  is the  $k^{\text{th}}$  term of this sequence,

- (b) find the value of  $S_{10}$  giving your answer to one decimal place.

(5)

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(Total for Question 8 is 7 marks)

(Total for Question 8 is 7 marks)

# Worked Solution - Question 8

## 1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

## 2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

## 3. State the final result

Write the answer in the form requested by the question: = -

### Final answer

= -

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12 marks

## Question 9

Circles

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

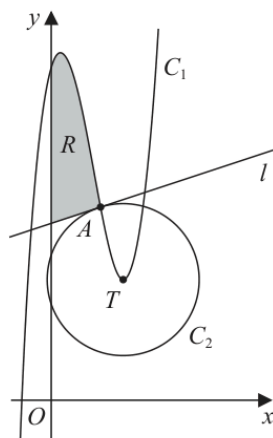


Figure 3

Figure 3 shows

- the curve  $C_1$  with equation  $y = x^3 - 5x^2 + 3x + 14$
- the circle  $C_2$  with centre  $T$

The point  $T$  is the minimum turning point of  $C_1$ 

Using Figure 3 and calculus,

- (a) find the coordinates of
- $T$

(3)

The curve  $C_1$  intersects the circle  $C_2$  at the point  $A$  with  $x$  coordinate 2

- (b) Find an equation of the circle
- $C_2$

(3)

The line  $l$  shown in Figure 3, is the tangent to circle  $C_2$  at  $A$ 

- (c) Show that an equation of
- $l$
- is

$$y = \frac{1}{3}x + \frac{22}{3}$$

(3)

The region  $R$ , shown shaded in Figure 3, is bounded by  $C_1$ ,  $l$  and the  $y$ -axis.

- (d) Find the exact area of
- $R$
- .

(3)

(Total for Question 9 is 12 marks)

(Total for Question 9 is 12 marks)

## Worked Solution - Question 9

### 1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

### 2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

### 3. State the final result

Write the answer in the form requested by the question: The required result is shown.

#### Final answer

The required result is shown.

## Question 10

## Laws of Logarithms

10. Given  $a = \log_2 3$

(i) write, in simplest form, in terms of  $a$ ,

(a)  $\log_2 9$

(b)  $\log_2 \left( \frac{\sqrt{3}}{16} \right)$

(3)

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of  $a$ .

(4)

(Total for Question 10 is 7 marks)

(Total for Question 10 is 7 marks)

# Worked Solution - Question 10

## 1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

## 2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

## 3. State the final result

Write the answer in the form requested by the question:  $- + = -- = 2 \ 2 \log \log b \ a$   
 $b \ a = 3 \ 1 \ a \ x \ a - = +$

### Final answer

$$- + = -- = 2 \ 2 \log \log b \ a \ b \ a = 3 \ 1 \ a \ x \ a - = +$$